Finite Excess in Ordinal Nim Multiplication: Four Selected-Order Problems

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Abstract

Lenstra's construction of ordinal nim multiplication assigns a finite excess m_p to every odd prime p. The observed values obey a simple 0/1/4 rule determined by the component support of $f(p) = \mathrm{ord}_p(2)$, but no universal proof is known.

We give an exact status theorem. The rule is equivalent to four full-primary-order assertions for elements selected by the literal Conway tower: a synchronized structural norm, a marked cyclotomic unit, a Gaussian period in a norm-one torus, and a corrected conductor-five norm. These are lossless reductions, not hypotheses, and all four universal assertions remain open. The paper then isolates the arithmetic content of each coordinate. It constructs the primitive-support quotient and the Conway–Fermat packet recursion, identifies the ordinary Artin symbol, proves the cubic Singer recursion, a partition selected-order bound, and a sharp first-relation threshold, proves an ordinary section sieve and a Kummer-ratio packing sieve, and gives the exceptional Capelli criterion, a primitive-only palindromicity test, a one-third Frobenius compression, an intermediate-field packing sieve, and a level-dependent translate bound. The compression proves the exceptional arm whenever its current cyclotomic quotient is prime, and the packing sieve proves a selected composite large-factor range, while the remaining composite levels stay open. A final reciprocity theorem shows that Frobenius, character, and affine-holonomy transforms of one marked Kummer phase can only erase or re-encode that phase; they cannot evaluate it without an independent ancestry input.

The proof record distinguishes paper theorems, ordinary kernel-checked algebra, native-decided named finite certificates, source-backed data, and the still-open universal rule.

1 Introduction

Conway's nim addition and multiplication make the ordinals a proper class field of characteristic two. Its finite subfields are canonically embedded, and its first transfinite layers are assembled by explicit Kummer extensions [3, 8, 9]. Lenstra's description associates to each odd prime p an element

$$\alpha_p = \kappa_{f(p)} + m_p, \qquad f(p) = \mathrm{ord}_p(2),$$

where $\kappa_{f(p)}$ is determined by the tower and $m_p \in \mathbb{N}$ is the *finite excess.* The same structure appears in the description of the ordinal fields On_p [5].

Let $\mathcal{Q}(h)$ be Lenstra's component set for κ_h. The available data suggest the following rule.

Conjecture 1.1 (the 0/1/4 rule)**.** *For every odd prime p,*

$$m_p = \begin{cases} 0, & \mathcal{Q}(f(p)) \text{ is not a singleton odd prime power}, \\ 4, & f(p) = 2\cdot 3^k \text{ for some } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

The term *selected* is essential. A finite field contains many elements having the same degree, trace, norm, or cyclotomic support, whereas the Conway construction recursively marks one Frobenius orbit. The open content is the primary multiplicative order of that orbit. Computations are valuable for individual rows and as countermodels to overstrong lemmas, but they do not prove Conjecture 1.1.

Theorem A. *The* 0/1/4 *rule is equivalent to the conjunction of four selected-order assertions* Z, O, C, D *stated in Theorem 3.1. Each assertion is an exact Euler nonpower test for one Conway-selected coordinate and retains the full primary valuation. The reductions, the coordinate formulas, the selected Singer and partition bounds of Propositions 6.3 and 6.4, the cubic first-relation theorem of Proposition 6.5, and the exceptional compression and packing and translate bounds of Propositions 7.6, 7.7, and 7.8 are proved here. Corollary 5.4 settles the ordinary large-factor case, Proposition 5.5 enlarges that range polynomially in* r^a*; the compression settles the exceptional prime-complement case, and the intermediate-field sieve settles an exponential-in-*$\sqrt{3^k}$ *composite large-factor range. None of* Z, O, C, D *is proved universally; consequently the* 0/1/4 *rule and absolute boundedness of the finite excess remain open.*

This theorem is the organizing spine of the paper. The point is not to replace one conjecture by four opaque ones, but to remove every ambient-field ambiguity while preserving the one datum that coarse invariants lose: the Frobenius orbit selected recursively by the Conway tower.

After finite-field preliminaries, Section 3 gives the four-arm equivalence. Sections 4–7 develop one lossless selected coordinate for each arm. Section 8 proves the saturation boundary for reciprocity arguments built from one marked phase. Section 9 separates the stronger 0/1/4 rule from absolute boundedness, and Section 10 records the authority of every claim class used in the paper.

2 Finite-field preliminaries

For a finite algebraic nimber x, write

$$d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2].$$

The element κ_h is the sum of the selected components κ_q, $q \in \mathcal{Q}(h)$, and $h \mid d(\kappa_h)$. The excess has the equivalent form

$$m_p = \min\{m \geq 0 : \kappa_{f(p)} + m \text{ is not a } p\text{-th power in its defining finite field}\}.$$

Finite integers are identified with their canonical finite nimbers. We use standard facts about finite fields and Capelli composition [10].

Proposition 2.1 (exact power criterion)**.** *Let* p *be prime, let* $p \mid 2^E - 1$*, and let* $x \in \mathbb{F}_{2^E}^{\times}$*. The following are equivalent:*

1. x *is a* p*-th power;*
2. $x^{(2^E-1)/p} = 1$*;*
3. $\operatorname{ord}(x) \mid (2^E - 1)/p$*.*

Consequently, x *is not a* p*-th power if and only if*

$$v_p(\operatorname{ord}(x)) = v_p(2^E - 1).$$

Proof. Choose a generator g of the cyclic multiplicative group and write $x = g^a$. Solvability of $py \equiv a \pmod{2^E - 1}$, the displayed power test, and the order divisibility are equivalent. The final statement is their p-primary form. $\square$

The valuation cannot be replaced by the weaker assertion $p \mid \mathrm{ord}(x)$. If $h = f(p)$ and $h \mid E$, lifting the exponent gives

$$v_p(2^E - 1) = v_p(2^h - 1) + v_p(E/h).$$

Every order assertion below retains the full right-hand valuation.

Proposition 2.2 (norm reduction)**.** *Let $h \mid E$, let $p \mid 2^h - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then*

$$x^{(2^E-1)/p} = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x)^{(2^h-1)/p}.$$

Thus x is a p-th power in $\mathbb{F}_{2^E}$ if and only if its norm is a p-th power in $\mathbb{F}_{2^h}$.

Proof. The norm exponent is $(2^E - 1)/(2^h - 1)$. Multiply it by $(2^h - 1)/p$ and apply Proposition 2.1. $\square$

3 Exact four-arm reduction

For $h > 1$, put

$$\Pi_h = \prod_{\substack{\ell \text{ odd prime} \\ \mathrm{ord}_\ell(2) = h}} \ell^{v_\ell(2^h-1)}, \qquad T_h = \mathrm{N}_{\mathbb{F}_{2^{d(\kappa_h)}}/\mathbb{F}_{2^h}}(\kappa_h).$$

Thus Π_h is the full primary part supported at primes first occurring at residue degree h.

For an odd prime r, let $b_r = d(\alpha_r)$. Put $x_{r,0} = \alpha_r$, and for $a \geq 1$ put

$$x_{r,a} = \kappa_{r^a}.$$

For every $a \geq 0$, define

$$F_{r,a} = \mathbb{F}_{2^{r^a b_r}}, \qquad y_{r,a} = x_{r,a} + 1,$$

and for $a \geq 1$ define

$$G_{r,a} = F_{r,a}^\times / F_{r,a-1}^\times, \qquad L_{r,a} = |G_{r,a}| = \Phi_{r^a}(2^{b_r}).$$

For $k \geq 1$, set $h = 3^k$, let $\zeta_k = \kappa_{3^k}$, and put

$$\gamma_k = \zeta_k + \zeta_k^{-1} \in \mathbb{F}_{2^h}^\times.$$

Finally, let $q = 2^h$, let $\omega = \zeta_k^h \in \mathbb{F}_4$, and define

$$A_k = \zeta_k^2 + \zeta_k + \omega, \qquad M_k = A_k^{q-1}, \qquad \Psi_k = \frac{\Phi_{2\cdot 3^k}(2)}{3}.$$

Theorem 3.1 (four-arm reduction)**.** *Conjecture 1.1 is equivalent to the conjunction of the following four universal assertions.*

*Z***.** *If $\mathcal{Q}(h)$ is not a singleton odd prime power, then $\Pi_h \mid \mathrm{ord}(T_h)$.*

*O***.** *If $r \neq 3$ is odd, $a \geq 1$, and $p \neq r$ is a prime divisor of $L_{r,a}$, then*

$$v_p\Big(\mathrm{ord}(y_{r,a} F_{r,a-1}^\times)\Big) = v_p(L_{r,a}).$$

C. The element γ_k is primitive in $\mathbb{F}_{2^{3^k}}^\times$ for every $k \geq 1$.

D. One has $\Psi_k \mid \operatorname{ord}(M_k)$ for every $k \geq 1$.

Proof. Fix p and put $h = f(p)$. If $\mathcal{Q}(h)$ is not a singleton odd prime power, Proposition 4.1 below identifies $m_p = 0$ with the corresponding primary factor of Z.

If $\mathcal{Q}(h) = \{r^a\}$ with r odd, Lenstra's component formulas give $h = r^a s$ with $s \mid b_r$. Proposition 5.1 identifies the row primes with $p \neq r$ dividing $L_{r,a}$ and identifies $m_p = 1$ with assertion O.

When $r = 3$, one has $b_3 = 2$. The cases are $h = 3^a$ and $h = 2 \cdot 3^a$. Proposition 6.1 identifies the former with C, while Theorem 7.3 identifies the latter with D. These cases are disjoint and exhaustive. $\square$

Remark 3.2. Theorem 3.1 is an equivalence, not a proof of any of its four universal assertions. None of Z, O, C, D is presently known in full.

4 The zero arm: structural norms and Fermat packets

4.1 Synchronized structural norm

Proposition 4.1 (exact support criterion)**.** *Let $E = d(\kappa_h)$. For an odd prime p with $f(p) = h$,*

$$m_p = 0 \quad \iff \quad v_p(\operatorname{ord}(T_h)) = v_p(2^h - 1).$$

Consequently every such row has excess zero if and only if $\Pi_h \mid \operatorname{ord}(T_h)$.

Proof. By the defining excess formula, $m_p = 0$ exactly when κ_h is not a p-th power. Proposition 2.2 moves the test to T_h, and Proposition 2.1 gives the primary valuation. $\square$

When κ_h has several components, T_h is not determined by separate component norms. Their common embedding carries a synchronized Frobenius phase. The following quotient removes every proper-subfield ambiguity while retaining that phase.

Proposition 4.2 (cubical primitive-support coordinate)**.** *Let S be the set of prime divisors of h, let $M = \mathbb{F}_{2^h}$, and write $\varphi(x) = x^2$ and $\sigma_r = \varphi^{h/r}$. For $A \subseteq S$, put $\sigma_A = \prod_{r \in A} \sigma_r$, and define*

$$H_h = \prod_{r \in S} \mathbb{F}_{2^{h/r}}^\times, \qquad \Delta_h(x) = \frac{\prod_{|A| \text{ even}} \sigma_A(x)}{\prod_{|A| \text{ odd}} \sigma_A(x)}.$$

Then

$$\ker \Delta_h = H_h, \qquad |\operatorname{im} \Delta_h| = \Phi_h(2),$$

and Δ_h induces an isomorphism $M^\times / H_h \simeq \operatorname{im} \Delta_h$. Arm Z at degree h is equivalent to

$$\Pi_h \mid \operatorname{ord}(\Delta_h(T_h)).$$

Proof. The operator is the power map

$$x \longmapsto x^{\prod_{r \mid h,\ r \text{ prime}} (1 - 2^{h/r})}.$$

In the cyclic group of order $2^h - 1$, lifting the exponent prime by prime shows that its kernel has order $\operatorname{lcm}_{r \in S}(2^{h/r} - 1)$, exactly the order of H_h. The cyclotomic product formula gives quotient order $\Phi_h(2)$. A prime with residue degree h divides no proper-subfield order, so the quotient preserves its full primary valuation. Proposition 4.1 gives the last assertion. $\square$

The cubical coordinate is canonical, but nonidentity is weaker than full primary order. Moreover, an additive cube made from independently shifted components is not the synchronized norm of their selected sum. Thus neither linear disjointness nor a nonzero raw cross-ratio proves Z.

4.2 The singleton-even Conway–Fermat chain

The zero arm also contains the singleton-even chain. Put

$$h_n = 2^{n+1}, \qquad c_n = \kappa_{h_n} = 2^{2^n}, \qquad E_n = \mathbb{F}_{2^{h_n}}, \qquad F_n = 2^{2^n} + 1,$$

and $E_{-1} = \mathbb{F}_2$. The tower relation is

$$c_n^2 + c_n = \prod_{j<n} c_j.$$

Let

$$\delta_n = \operatorname{ord}(c_n E_{n-1}^\times) \quad \text{in } E_n^\times / E_{n-1}^\times.$$

Theorem 4.3 (Conway–Fermat selected order)**.** *One has $1 < \delta_n \mid F_n$, and the zero arm at every prime divisor of F_n is equivalent to*

$$\delta_n = F_n \qquad (n \geq 0).$$

All compatible binary root choices form one absolute-Frobenius orbit, so they have the same order vector $(\delta_0, \ldots, \delta_n)$. The remaining assertion is therefore intrinsic to the selected tower rather than to an extra choice of root.

Proof. The quotient group has order $(2^{2^{n+1}} - 1)/(2^{2^n} - 1) = F_n$, and $c_n \notin E_{n-1}$. Every prime $p \mid F_n$ has $f(p) = 2^{n+1}$, with its full primary part in this quotient. Proposition 2.1 gives the displayed equality for δ_n.

A compatible prefix has two extensions at each level, so there are 2^{n+1} compatible tuples. Frobenius preserves the defining equations, and the 2^{n+1} Frobenius conjugates are distinct because c_n has that degree. They exhaust the tuples and preserve all relative orders. □

The recurrence has a deterministic packet form. If $u_n = (c_n + 1)/c_n$, then $\operatorname{ord}(u_n) = \delta_n$ and $a_{n-1} = \prod_{j<n} c_j = u_n/(u_n + 1)^2$. For each divisor $e > 1$ of F_n, the inverse pairs of primitive e-th roots give a squarefree packet polynomial

$$C_e(X) = \prod_{[z] \in \mu_e^{\text{prim}}/(z \sim z^{-1})} \left(X - \frac{z}{(z+1)^2} \right) \in \mathbb{F}_2[X].$$

Every irreducible factor has degree 2^n, and

$$\delta_n = e \quad \iff \quad \text{the selected minimal polynomial of } a_{n-1} \text{ divides } C_e.$$

If P is such a factor, its unique irreducible child is

$$\operatorname{Res}_X(P(X), Y^2 + XY + X^3).$$

Thus the literal chain is one deterministic orbit beginning at $X + 1$. Full parent conductor does not imply full child conductor: exact degree-16 packet examples have identical full parent order but deterministic children of different conductor. Consequently any induction must use the complete marked orbit, not only the integer conductor.

The literal multiplication operator retains that selector in a compact two-block form. At the edge $c_n^2 = c_n + a_{n-1}$, let A denote multiplication by a_{n-1} on E_{n-1}. Multiplication by $a_{n-1}c_n$ on $E_{n-1} \oplus c_n E_{n-1}$ has matrix

$$\begin{pmatrix} 0 & A^2 \\ A & A \end{pmatrix}.$$

This identity is kernel-checked. The Lean declaration is

`fermat_selected_product_`
`mul_pair`.

If S_r denotes the characteristic-two Fibonacci polynomial, the genuine level-n target is

$$S_{F_n/\ell}(a_{n-1}) \neq 0 \qquad (\ell \mid F_n \text{ prime}).$$

Equivalently, one applies the same polynomial to multiplication by a_{n-1} on E_{n-1}. Write M_n for the displayed two-block operator. It represents multiplication by $a_n = a_{n-1}c_n$, so it is the recursive interface to the *next*-level target

$$S_{F_{n+1}/\ell}(M_n) \neq 0 \qquad (\ell \mid F_{n+1} \text{ prime}).$$

At the lower index one instead has the automatic exclusion

$$S_{F_n/\ell}(a_n) \neq 0, \quad \text{equivalently} \quad S_{F_n/\ell}(M_n) \neq 0 \qquad (\ell \mid F_n),$$

because a zero would put a_n in E_{n-1}: zero-block propagation reaches S_{F_n}, and the telescoping partial-trace identity then forces $a_n^{|E_{n-1}|} = a_n$, whereas its Frobenius displacement is $a_{n-1} \neq 0$. This wrong-index exclusion is kernel-checked as

`fibPolyValue_quadratic_birth_`
`lower_period_ne_zero`.

It is not the selected-order conjecture. The block formula remains an exact interface to the displayed next-level nonvanishing problem; it does not prove that target.

At the genuine index there is nevertheless a sharper exact residual. Let $d = F_{n+1}/\ell = 2m+1 > 1$ and put $x = a_n$. Characteristic-two doubling and Cassini give

$$S_m(x)^4 + x^{d-2} = S_d(x)S_{d-2}(x),$$
$$S_{m+1}(x)^4 + x^d = S_d(x)S_{d+2}(x).$$

Consequently, over a field,

$$S_d(x) = 0 \quad \iff \quad S_m(x)^4 = x^{d-2} \qquad \text{and} \qquad S_{m+1}(x)^4 = x^d.$$

The converse is literal: squaring $S_d = S_{m+1}^2 + xS_m^2$ makes the two displayed powers cancel. The pair is strictly sharper than either equation separately. Indeed, in $\mathbb{F}_2[X]$ its two left-hand residual polynomials have monic gcd exactly S_d. To see the only needed coprimality, any common divisor of S_{d-2} and S_{d+2} divides $X^{d-2}S_4 = X^{d-2}$ by the continuant determinant identity, while the positive odd-index polynomial S_{d-2} has constant coefficient one. Thus S_{d-2} and S_{d+2} are coprime, and the two factorizations give the stated gcd.

This exact branch selector still does not close the Conway–Fermat target. If $d-1 = 2^s g$ with g odd, its lower equation is

$$S_g(x)^{2^{s+1}} = x^{d-2}.$$

Writing $q = |E_n|$, choose K by the exact congruence $2^{s+1}K \equiv d-2 \pmod{q-1}$. Since $x \neq 0$, $x^{q-1} = 1$, and the 2^{s+1}-power Frobenius is injective, the equation becomes $S_g(x) = x^K$. This is precisely the existing AE4 inverse-exponent seam. The upper equation removes the adjacent Fibonacci branch but, after $S_d(x) = 0$ is imposed, follows by squaring the same endpoint relation twice and using the lower equation; no present ancestry identity contradicts the resulting monomial equality. The two ordinary identities and the field equivalence are kernel-checked as

`fibPolyValue_odd_lower_fourth_factor`,
`fibPolyValue_odd_upper_fourth_factor`, and
`fibPolyValue_odd_eq_zero_iff_fourth_residuals`.

This isolates the part not settled by the existing Conway-tower literature. Popovych proves lower bounds and verifies the required relative full orders through level eleven using the known Fermat factorizations [13]. Cagliero, Herman, and Szechtman prove an exact product formula for global order in terms of the relative orders, as well as a normal basis theorem for the associated tower elements [2]. Neither result forces every relative factor δ_n to equal F_n; that all-level equality is precisely the remaining selected assertion here.

There is an exact supersingular realization of the selected ancestry.

Proposition 4.4 (supersingular realization and pairing boundary)**.** *On*

$$\mathcal{E}: \quad y^2 + y = x^3 + x^2$$

put $P_n = (c_n, c_{n+1})$. Then $P_n \in \mathcal{E}$, and for $n \geq 2$,

$$[F_n]P_n = O.$$

More precisely, if π is absolute Frobenius and $\iota = \pi + [1]$, then

$$\iota(x,y) = (x+1, y+x+1), \qquad \iota^2 = [-1], \qquad \pi^4 = [-4].$$

For $g = (y+1)/y$, one has

$$g(P_n) = \frac{c_{n+1}+1}{c_{n+1}}, \qquad \operatorname{ord}(g(P_n)) = \delta_{n+1}.$$

If $T = (0,0)$, then T has order five and

$$\operatorname{div}(g) = 2[-T] + [2T] - 2[T] - [-2T].$$

Thus g is the quotient of the two fixed doubling Miller functions at $-T$ and T, but it is not a Weil or reduced-Tate pairing function attached to the F_n-torsion point P_n when $n \geq 2$.

Proof. The Conway recursion gives

$$c_{n+1}^2 + c_{n+1} = c_n(c_n^2 + c_n) = c_n^3 + c_n^2,$$

so P_n lies on $\mathcal{E}$. The coordinate formula for $\iota(P) = \pi(P) + P$ follows from the chord law. Applying it twice gives the inverse $(x,y) \mapsto (x, y+1)$, so $\pi^2 + 2\pi + 2 = 0$ in $\operatorname{End}(\mathcal{E})$ and hence $\pi^4 = [-4]$.

Put $H = 2^{n+1}$. The 2^H-Frobenius fixes c_n and sends c_{n+1} to $c_{n+1}+1$, hence sends P_n to $-P_n$. For $n \geq 2$,

$$\pi^H = (\pi^4)^{2^{n-1}} = [2^{2^n}],$$

which gives $[2^{2^n}+1]P_n = O$.

The formula for $g(P_n)$ is immediate from the coordinates, and its order is δ_{n+1} by the definition of u_{n+1}. The five rational points of $\mathcal{E}/\mathbb{F}_2$ show that T has order five. Direct intersection with the two horizontal lines gives

$$\mathrm{div}(y) = 2[T] + [-2T] - 3[O], \qquad \mathrm{div}(y+1) = 2[-T] + [2T] - 3[O],$$

and subtraction gives the divisor of g. Equivalently, $g = f_{2,-T}/f_{2,T}$ for functions normalized at O. Its divisor is supported on fixed rational five-torsion, with coefficients $2, 1, -2, -1$. For $n \geq 2$, these valuations are not divisible by $F_n \geq 17$, whereas the divisor of an F_n-Miller function and every F_n-th-power correction have all coefficients divisible by F_n. Thus g cannot be such a function attached to P_n, even modulo an F_n-th power. A Weil or reduced Tate pairing on $\mathcal{E}[F_n]$ takes values in μ_{F_n}, whereas $g(P_n)$ has order dividing F_{n+1}. Pairwise coprimality of the Fermat numbers and $\delta_{n+1} > 1$ exclude equality. □

The proposition supplies exact CM annihilator control but not generator status. For $n \geq 3$, no elliptic endomorphism can carry P_n to P_{n-1}: the image order would divide both coprime numbers F_n and F_{n-1}, although $P_{n-1} \neq O$. For $n = 2$, the same conclusion follows from $\mathrm{ord}(P_1) = 3$ and $\mathrm{ord}(P_2) \mid 17$. Thus the relation $x(P_n) = y(P_{n-1})$ is not an isogeny division tower, and pairing functoriality along this proposed tower cannot propagate primitivity. A perfect alternating pairing likewise guarantees some detecting partner, not nonvanishing on a prescribed line.

The tempting cubic curve $Y^2 + Y = X^3$ is not a coordinate rewrite of $\mathcal{E}$: the two curves have respectively three and five rational points over $\mathbb{F}_2$. For $u = (Y+1)/Y$ on that auxiliary curve, $u/(u+1)^2 = X^3$; every positive-level Fermat-torsion value is already a cube because $\gcd(F_n, 3) = 1$. Its natural cubic Kummer class is therefore trivial and cannot replace the selected five-torsion function above. The remaining statement is still exclusion of every proper-conductor packet along the complete resultant chain. Trace, relative norm, irreducibility, and the etale Jacobian hold for proper packets as well as the full packet.

5 The ordinary arm: a marked cyclotomic unit

Fix an odd prime r, write $b = b_r$, and abbreviate $x_a = x_{r,a}$, $y_a = x_a + 1$, and $F_a = F_{r,a}$.

Proposition 5.1 (projective order criterion)**.** *For $a \geq 1$,*

$$\mathrm{N}_{F_a/F_{a-1}}(y_a) = y_{a-1}, \qquad |F_a^\times / F_{a-1}^\times| = L_{r,a}.$$

If $p \neq r$ divides $L_{r,a}$, then the corresponding singleton row has

$$m_p = 1 \quad \Longleftrightarrow \quad v_p\big(\mathrm{ord}(y_a F_{a-1}^\times)\big) = v_p(L_{r,a}).$$

Every singleton row with component r^a arises this way.

Proof. The relative conjugates of x_a are ξx_a for $\xi \in \mu_r$; evaluating their binomial at one gives the norm identity. The quotient order is the quotient of the two finite-field unit orders. The cyclotomic divisor criterion gives $\mathrm{ord}_p(2^b) = r^a$ for $p \neq r$ dividing $L_{r,a}$, and Lenstra's component theorem identifies precisely the singleton rows. The lower unit group has no p-part, so Proposition 2.1 gives the last equivalence. □

The full recursive ancestry can also be retained in one binary polynomial. Let $P_r \in \mathbb{F}_2[T]$ be the minimal polynomial of $x_0 = \alpha_r$.

Proposition 5.2 (bottom-ancestor divisibility)**.** *Put* $q = |F_{a-1}|$*,* $N = L_{r,a}$*, and let* $p \neq r$ *divide* N*. Then* $\gcd(p, q-1) = 1$*. Write* $d = N/p$*, and choose* $e, t \geq 0$ *such that*

$$pe = 1 + t(q-1).$$

Then the current p*-coordinate of* $y_a F_{a-1}^{\times}$ *is deficient if and only if*

$$P_r(T^{r^a}) \mid (1+T)^d + (1+T^r)^e \qquad \textit{in } \mathbb{F}_2[T].$$

Proof. Indeed, $N \equiv r \pmod{q-1}$, so $\gcd(N, q-1) = r$. The polynomial $P_r(T^{r^a})$ is the minimal polynomial of x_a. The coordinate is deficient exactly when $c = (1+x_a)^d$ lies in $F_{a-1}^{\times}$. In that case the norm identity gives

$$c^p = (1+x_a)^N = 1 + x_a^r.$$

Since $c^{q-1} = 1$ and $pe = 1 + t(q-1)$, $c = (c^p)^e = (1+x_a^r)^e$. Conversely this equality puts c in F_{a-1}. In characteristic two the equality is precisely the vanishing at x_a of the displayed binary polynomial, which is equivalent to divisibility by its minimal polynomial. □

This is a lossless selector-retaining reduction, not a proof of arm O: the remaining assertion is that the displayed divisibility never occurs for the literal polynomial P_r and every current factor p.

For a polynomial $B(T) = \sum_m b_m T^m$ and $0 \leq j < n$, define its j-th n-section by

$$\mathrm{Sec}_{n,j}(B)(U) = \sum_{k \geq 0} b_{j+kn} U^k.$$

Lemma 5.3 (polynomial section criterion)**.** *Let* k *be a field, let* $0 \neq P \in k[U]$*, and let* $n \geq 1$*. Then*

$$P(T^n) \mid B(T) \iff P(U) \mid \mathrm{Sec}_{n,j}(B)(U) \quad (0 \leq j < n).$$

Proof. Every polynomial has the unique decomposition $B(T) = \sum_{j<n} T^j \,\mathrm{Sec}_{n,j}(B)(T^n)$. Multiplication by $P(T^n)$ preserves each residue class of exponents modulo n. Applying the decomposition to the quotient proves the forward implication, and summing the sectionwise quotients proves the converse. □

Corollary 5.4 (Lucas-section ordinary case)**.** *Put* $n = r^a$*, let* $p \neq r$ *divide* $N = L_{r,a}$*, and put* $d = N/p$*. If some* $0 \leq j < n$ *satisfies*

$$r \nmid j, \qquad \binom{d}{j} \equiv 1 \pmod 2, \qquad d - j < bn,$$

then the current p*-coordinate of* $y_a F_{a-1}^{\times}$ *has full primary order. In particular this holds if* $d \leq bn$*, and hence when* N/r *is a prime distinct from* r*.*

Proof. In the polynomial of Proposition 5.2, the j-th section receives no term from $(1+T^r)^e$, since $r \mid n$ but $r \nmid j$. Its other summand has constant coefficient one and degree at most $\lfloor (d-j)/n \rfloor < b$. It is therefore a nonzero polynomial of degree less than $\deg P_r = b$, so Lemma 5.3 rules out the required divisibility. Finally, d is odd, so when $d \leq bn$ one may take $j = 1$. If $p = N/r$, then $d = r \leq br^a$. □

The full Kummer orbit gives a second sieve which is substantially stronger when the tower level is large.

Proposition 5.5 (Kummer-ratio packing sieve)**.** *Assume $r \neq 3$ and $a \geq 1$, put $n = r^a$, and let the prime $p \neq r$ divide $N = L_{r,a}$. Set*

$$d = N/p, \qquad D = r^{a-1}(r-1), \qquad s = (r-1)/2.$$

If the current p-coordinate is deficient, then

$$d \geq \mathcal{R}(D,s) + \left\lceil \frac{\mathcal{R}(D,s+1) - \mathcal{R}(D,s)}{r} \right\rceil, \qquad \mathcal{R}(D,t) := \sum_{j=0}^{t} 2^j \binom{D}{j}\binom{t}{j}.$$

Consequently arm O holds at every such factor satisfying

$$\frac{L_{r,a}}{p} < \mathcal{R}\left(r^{a-1}(r-1), \frac{r-1}{2}\right) + \left\lceil \frac{1}{r}\left[\mathcal{R}\left(r^{a-1}(r-1), \frac{r+1}{2}\right) - \mathcal{R}\left(r^{a-1}(r-1), \frac{r-1}{2}\right)\right]\right\rceil.$$

Proof. Put $K = F_{a-1}$ and $x = x_a$. The nonpower radicand gives $r \mid 2^b - 1$, so LTE gives $r^a \mid 2^{br^{a-1}} - 1$; hence K contains μ_n. Since $F_a = K(x)$, the element x has minimal-polynomial degree r over K. Choose representatives $\mathcal{C}$ for μ_n/μ_r. Deficiency and Proposition 5.2 give, at every root cx of $P_r(T^n)$ and every $\zeta \in \mu_r$,

$$(1+\zeta cx)^d = (1+(\zeta cx)^r)^e = (1+(cx)^r)^e = (1+cx)^d.$$

Every numerator and denominator below is nonzero: an equality $1+\xi x = 0$ with $\xi \in \mu_n$ would put x in K, contradicting its relative degree $r > 1$. Thus each of the

$$u_{c,\zeta} = \frac{1+\zeta cx}{1+cx}, \qquad c \in \mathcal{C}, \quad \zeta \in \mu_r \setminus \{1\},$$

is a d-th root of unity. There are $|\mathcal{C}|(r-1) = D$ such ratios.

Index these ratios by a set I of size D. For every vector $v = (v_i)_{i\in I} \in \mathbb{Z}^I$ with $\sum_i |v_i| \leq s$, put $U_v = \prod_i u_i^{v_i}$. We claim that the U_v are distinct. If $U_v = U_w$, set $g_i = v_i - w_i$. After separating positive and negative exponents, the resulting rational-function equality at x has the form

$$A(x) = B(x), \qquad \deg A, \deg B \leq \sum_i |g_i| \leq 2s = r-1 < r,$$

where $A, B \in K[T]$ have constant coefficient one. Since the minimal polynomial of x over K has degree r, one has $A = B$ as polynomials. This low-degree step is the ordinary Lean theorem `polynomial_eq_of_aeval_eq_of_natDegree_lt_minpoly`.

For a fixed $c \in \mathcal{C}$, the ratios form the star

$$\frac{1+\zeta cT}{1+cT}, \qquad \zeta \in \mu_r \setminus \{1\}.$$

Stars belonging to different cosets have disjoint linear factors. Within a star, the valuation at the distinct factor $1+\zeta cT$ recovers the corresponding exponent. Unique factorization therefore forces every $g_i = 0$, and hence $v = w$.

To count these integer vectors, choose the j nonzero coordinates, their signs, and their positive absolute values. There are $\binom{D}{j}$, 2^j, and $\binom{s}{j}$ choices respectively, the last count being the number of positive j-tuples with sum at most s. Hence there are exactly $\mathcal{R}(D,s)$ vectors. Their values are distinct roots of X^d-1, which has at most d roots in a field. This proves the inner-ball contribution.

Part of the next signed shell can be retained after separating the ratios by their leading phase. For $i=(c,\zeta)$ put $\chi_i=\zeta$ and, for $v\in\mathbb{Z}^I$, put

$$\chi(v)=\prod_i \chi_i^{v_i}\in\mu_r.$$

The outer shell of vectors of signed ℓ^1-weight exactly $s+1$ has $\mathcal{R}(D,s+1)-\mathcal{R}(D,s)$ elements. Some fibre H of χ in that shell therefore has at least

$$\left\lceil \frac{\mathcal{R}(D,s+1)-\mathcal{R}(D,s)}{r}\right\rceil$$

elements. We claim that the U_v for the entire inner ball together with H are distinct.

Indeed, suppose that $U_v=U_w$ for two vectors in this union and set $g=v-w$. Form the same cross-products $A,B\in K[T]$, without cancelling common factors. Both have constant coefficient one and exact degree

$$t=\sum_i |g_i|.$$

If at least one vector is in the inner ball, then $t\le s+(s+1)=r$. If both lie in H, then $t\le 2(s+1)=r+1$, while the quotient of the leading coefficients is $\prod_i \chi_i^{g_i}=\chi(v)/\chi(w)=1$; their leading terms cancel. Thus in either case $A-B=0$ or $\deg(A-B)\le r$. The minimal polynomial m_x divides $A-B$. If the difference were nonzero, this degree bound and $\deg m_x=r$ would give $A-B=\lambda m_x$ for some $\lambda\in K^\times$. But $A-B$ has constant coefficient zero, whereas the relative Kummer minimal polynomial m_x has nonzero constant coefficient, a contradiction. This boundary step is the ordinary Lean theorem `polynomial_eq_of_aeval_eq_of_natDegree_le_minpoly_of_coeff_zero`. Thus $A=B$, and the same disjoint-star valuations force $g=0$.

The union therefore injects into the roots of X^d-1. Its cardinality is at least the displayed inner-ball-plus-shell expression, proving the bound and its contrapositive. □

Let $N_a=\operatorname{ord}(x_a)$, choose a primitive complex N_a-th root ξ_{N_a}, and let $\mathfrak{P}_a\mid 2$ be the prime whose residue sends ξ_{N_a} to x_a. Put $\varepsilon_a=1-\xi_{N_a}$.

Theorem 5.6 (selected cyclotomic-unit form)**.** *For $p\neq r$ dividing $L_{r,a}$, arm O at (r,a,p) is equivalent to*

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a}\right)_p\neq 1.$$

The compatible primes $\mathfrak{P}_a$ form a unique inert ray above two: $\mathfrak{P}_a$ is the unique prime above $\mathfrak{P}_{a-1}$, with relative residue degree r, and

$$\mathrm{N}_{\mathbb{Q}(\xi_{N_a})/\mathbb{Q}(\xi_{N_{a-1}})}(1-\xi_{N_a})=1-\xi_{N_{a-1}}.$$

Nevertheless, residue-field corestriction kills the new p-coordinate and does not determine the displayed residue symbol.

Proof. The reduction of ε_a is y_a. The unramified power-residue formula gives

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a}\right)_p=y_a^{(|F_a|-1)/p},$$

so Proposition 2.1 and Proposition 5.1 give the equivalence.

The Kummer steps increase the r-primary part of N_a by one and give $\mathrm{ord}_{N_a}(2) = br^a$. The relative cyclotomic degree and the residue-degree ratio are therefore both r, forcing one prime above the preceding selected prime. The global norm identity is the product over the r-th roots of unity. Finally, $p \mid \Phi_{r^a}(2^b)$ but $p \nmid 2^{br^{a-1}} - 1$, so the lower residue field has trivial p-Kummer quotient. Its norm receives no information about the new selected class. $\square$

The selected symbol has an exact Jacobi–Fourier expansion. Let $Q = |F_a|$, let $H = \langle x_a \rangle \subset F_a^\times$ have order N_a, put $M = (Q-1)/N_a$, and let χ be a complex-valued order-p power character. Extend complex-valued multiplicative characters by zero at zero and define

$$f(h) = \chi(1-h), \qquad \widehat{f}(\theta) = \sum_{h \in H} f(h)\theta(h)^{-1} \quad (\theta \in \widehat{H}).$$

For an extension ψ_θ of θ to $F_a^\times$, and with $H^\perp$ the character group of $F_a^\times/H$, one has

$$\widehat{f}(\theta) = \frac{1}{M} \sum_{\lambda \in H^\perp} J(\psi_\theta^{-1}\lambda, \chi), \qquad f(x_a) = \frac{1}{N_a} \sum_{\theta \in \widehat{H}} \theta(x_a)\widehat{f}(\theta).$$

Indeed, the character average $M^{-1}\sum_{\lambda \in H^\perp} \lambda$ is the indicator of H; inserting it in the first Fourier sum gives the Jacobi sums, and Fourier inversion gives the second identity. This is a fully explicit reciprocity formula, but the selected label survives in $\theta(x_a)$ and the Jacobi unit phases. Ideal factorization or valuation formulas for the Jacobi sums do not by themselves determine this inverse-DFT entry. This separation of the ideal and p-adic unit conditions is also visible in Uehara's cyclotomic-unit criterion [16]; it supplies an exact test, not a uniform nonvanishing theorem for the Conway-selected entry.

This residue-symbol condition is a marked Artin-symbol assertion, not a statement about decomposition or conductor. Its ray character splits into a ramified local logarithm and an unramified ideal-class character evaluated at $[\mathfrak{P}_a]$. Stickelberger annihilation reduces the latter to an explicit decomposition-invariant quotient, but does not make it vanish uniformly. This is a genuine boundary: at the actual Conway conductor $r = 11$, $a = 1$, $p = 23$, one contributing Stickelberger scalar vanishes with exact 23-adic valuation one. The corresponding unramified degree-23 class line is nontrivial, and every prime over two has nontrivial Artin symbol on it, although the literal row still has $m_{23} = 1$. Thus successful rows need not have a zero unramified term; the open assertion is noncancellation in the actual selected sum.

6 The cubic arm: a selected Singer generator

Let $h = 3^k$, $\zeta_k = \kappa_{3^k}$, and $\gamma_k = \zeta_k + \zeta_k^{-1}$.

Proposition 6.1 (half-angle reduction)**.** *The element ζ_k is a primitive 3^{k+1}-st root of unity,*

$$\mathbb{F}_2(\zeta_k) = \mathbb{F}_{2^{2h}}, \qquad \mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}.$$

Moreover,

$$\mathrm{ord}(\zeta_k + 1) = 3^{k+1}\,\mathrm{ord}(\gamma_k),$$

and the translates by 2 and 3 have the same order. Hence the rows with $f(p) = 3^k$ have excess one precisely when γ_k has the full relevant primary order; over all levels this is equivalent to primitivity of every γ_k.

Proof. The defining Kummer equation gives $\zeta_k^{3^k} = \kappa_2$, a primitive cube root. The order of two modulo 3^{k+1} is $2h$, and the least exponent giving ± 1 is h, proving the field degrees. If s is the inverse of two modulo 3^{k+1}, then

$$\zeta_k + 1 = \zeta_k^s(\zeta_k^s + \zeta_k^{-s}), \qquad (\zeta_k^s + \zeta_k^{-s})^2 = \gamma_k.$$

The factors lie in the coprime circle and real parts of the unit group, giving the order identity. Frobenius gives the other translates, and Proposition 2.1 finishes the reduction. $\square$

The half-angle test also has a lossless group-algebra form that removes all denominators. Put

$$J_h(X) = X^{2h} + X^h + 1, \qquad B_d(X) = (X^2+1)^d + X^d \in \mathbb{F}_2[X].$$

Since J_h is the minimal polynomial of ζ_k and $\gamma_k = \zeta_k^{-1}(\zeta_k^2 + 1)$, one has

$$\gamma_k^d = 1 \iff J_h(X) \mid B_d(X). \tag{1}$$

Equivalently, in $\mathbb{F}_2[X, X^{-1}]/(X^{3h} - 1)$ the parity vector of $X^d(1+X^{-2})^d$ belongs to the affine coset $1 + (J_h)$, not to the ideal (J_h) itself. Thus arm C is exactly the universal assertion that, for every prime $\ell \mid 2^h - 1$,

$$J_h(X) \nmid B_{(2^h-1)/\ell}(X).$$

Lucas's theorem makes the support of B_d explicit from the binary digits of d, but it does not by itself prove these selected nondivisibilities. The denominator-free equivalence above is also kernel-checked as `halfAngle_pow_eq_one_iff_minpoly_dvd`; it is a reduction of C, not a proof of C.

There is also an exact stable-polynomial formulation. Put $P_0(X) = X$ and

$$P_{j+1}(X) = P_j(X)^3 + P_j(X)^2 + 1,$$

so that $P_j = f^{\circ j}$ for $f(X) = X^3 + X^2 + 1$. In characteristic two, $t(X) = X + 1$ conjugates $g(X) = X^3 + X$ to f: $f = t \circ g \circ t$. The triple-angle identity gives $g(\gamma_j) = \gamma_{j-1}$ with $\gamma_0 = 1$, hence $b_j = \gamma_j + 1$ is a root of P_j. Since b_j has degree 3^j, P_j is its minimal polynomial. Lin and Wang prove that f is stable, equivalently that all these iterates are irreducible [11]; their theorem does not assert multiplicative primitivity.

Indeed, this missing strengthening is exactly arm C. With $q = 2^{3^{j-1}}$, both γ_j and b_j have relative norm γ_{j-1} to $\mathbb{F}_q$, and

$$\gamma_j b_j^2 = \gamma_{j-1}.$$

Because $\gcd(q-1, q^2+q+1) = 1$, the norm propagates every old primary part. For a new prime $\ell \mid q^2 + q + 1$, the base-field factor γ_{j-1} is an ℓ-th power, while the displayed identity gives $[\gamma_j] + 2[b_j] = 0$ in the ℓ-Kummer quotient. Since ℓ is odd, γ_j is an ℓ-th power exactly when b_j is. Induction therefore shows that all P_j with $j \geq 1$ are primitive polynomials exactly when all γ_j with $j \geq 1$ are primitive. Stability supplies the degrees; primitive preservation is the original selected-order problem, not a known solution of it.

Put $q_k = 2^{3^{k-1}}$ and

$$U_k = \ker\left(\mathrm{N}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}\right), \qquad \eta_k = \gamma_k^{q_k - 1}, \qquad \eta_0 = 0.$$

Theorem 6.2 (selected norm-one coordinate)**.** *One has $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$. The element η_k generates $\mathbb{F}_{2^{3^k}}$ and is the selected root of*

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1.$$

Furthermore,

$$C_k \iff C_{k-1} \text{ and } \operatorname{ord}(\eta_k) = |U_k|,$$

where C_j denotes primitivity of γ_j.

Proof. The definition puts η_k in the norm-one group. Frobenius calculation from ζ_k gives the displayed cubic; its coefficients are respectively the trace, second symmetric function, and norm. A base-field element of U_k has order dividing $\gcd(q_k - 1, 3) = 1$, so the nontrivial cubic is irreducible. Its quadratic coefficient recursively recovers η_{k-1}, proving the field degree. Finally, the finite-field unit group splits into its base-field and norm-one factors, whose orders are coprime. The norm propagates the old part of γ_k and η_k detects exactly the new factor. $\square$

The recursion selects one Frobenius orbit inside a Singer difference set. Indeed, for

$$R(x) = \frac{x^3 + x + 1}{x^2 + x}, \qquad W(x) = \frac{x + \omega}{x + \omega^2},$$

one has

$$R(\eta_k) = \eta_{k-1}, \qquad W(R(x)) = W(x)^3, \qquad W(\eta_k) = \zeta_k^2.$$

Thus the complete lower ancestry, not merely degree or trace, marks the orbit. The displayed conjugacy exposes its cyclotomic phase but does not evaluate the multiplicative order of η_k.

Proposition 6.3 (selected Singer sieve)**.** *Let $N = |U_k|$ and $h = 3^k$. Then*

$$\operatorname{ord}(\eta_k) \geq h(h-1) + 1.$$

If a prime $\ell \mid N$ is missing from the selected order at its full valuation, then

$$\frac{N}{\ell} \geq h(h-1) + 1.$$

Hence every prime $\ell > N/(h(h-1)+1)$ occurs with full valuation.

Proof. The selected Frobenius orbit has h points in the Singer $(N, q_k + 1, 1)$ difference set. Its $h(h-1)$ ordered off-diagonal quotients are distinct nonidentity elements of $\langle \eta_k \rangle$, giving the first bound. If the full ℓ-part is missing, the orbit lies in the subgroup of order N/ℓ, which must contain all these quotients. $\square$

The sieve is polynomial in h, while N is exponential. It proves a genuine selected large-factor range but not arm C.

The cyclotomic selector also permits a complementary partition bound. This is the usual low-degree collision argument for Gaussian periods, specialized to the composite conductor 3^{k+1} and therefore retaining the literal selected element rather than replacing it by an arbitrary normal generator.

Proposition 6.4 (selected norm-one partition bound)**.** *Let $n = 3^{k+1} = 3h$. For an integer $m \geq 1$ put*

$$\mathcal{A}_m = \{1 \leq j \leq m : 3 \nmid j\}, \qquad t_m = |\mathcal{A}_m|, \qquad s_m = \sum_{j \in \mathcal{A}_m} j.$$

Let

$$\mathcal{U}_h = \{1 \leq j < n : 3 \nmid j\},$$

choose $0 \le z_j < 2h$ *with* $2^{z_j} \equiv j \pmod n$*, and, for* $E \subseteq \mathcal{U}_h$*, put*

$$Q_E = \sum_{j\in E} 2^{z_j}, \qquad s_E = \sum_{j\in E} j.$$

For distinct $E, F \subseteq \mathcal{U}_h$*, set*

$$A = E \setminus F, \quad B = F \setminus E, \quad W = s_A + s_B, \quad j_0 = \min(A \cup B).$$

Then $\eta_k^{Q_E} \ne \eta_k^{Q_F}$ *if either*

$$s_A \not\equiv s_B \pmod 3 \ \text{ and } \ W < h, \qquad \text{or} \qquad s_A \equiv s_B \pmod 3 \ \text{ and } \ W - 2j_0 < h.$$

In particular, weighted symmetric-difference sum below h *rules out every collision.*

Put

$$R_h = \frac{h-1}{2}, \qquad \mathcal{P}_h = \sum_{r=0}^{R_h} [X^r] \prod_{\substack{j\ge 1\\ 3\nmid j}} (1 + X^j).$$

Then

$$\operatorname{ord}(\eta_k) \ge \mathcal{P}_h, \qquad \log_2 \mathcal{P}_h = \left(\frac{\pi}{3\log 2} + o(1)\right)\sqrt{h}.$$

If $s_m < h$*, then also*

$$\operatorname{ord}(\eta_k) \ge 2^{t_m}.$$

Consequently

$$\log_2 \operatorname{ord}(\eta_k) \ge \left(\frac{\pi}{3\log 2} + o(1)\right)\sqrt{h},$$

and the elementary initial-segment subfamily alone gives

$$\log_2 \operatorname{ord}(\eta_k) \ge \sqrt{\frac{4h}{3}} - O(1).$$

If a prime $\ell \mid N = |U_k|$ *is absent from the order of* η_k *at its full valuation, then* $N/\ell \ge \mathcal{P}_h$ *and, whenever* $s_m < h$*, also* $N/\ell \ge 2^{t_m}$*.*

Proof. Put $w = \zeta_k^2$. The identity $W(\eta_k) = w$ gives

$$\eta_k = \frac{\omega + \omega^2 w}{1 + w}.$$

The order of two modulo n is $2h$, so the exponents z_j in the statement exist. They satisfy

$$\eta_k^{2^{z_j}} = \omega^j \frac{1 + (\omega w)^j}{1 + w^j}.$$

A collision $\eta_k^{Q_E} = \eta_k^{Q_F}$ first permits cancellation of all indices in $E \cap F$: the relevant numerator and denominator factors are nonzero at w, because both w and ωw are primitive n-th roots. Put $A = E \setminus F$, $B = F \setminus E$, and $W = s_A + s_B$ as in the statement. Clearing the remaining denominators makes w a root of the difference between

$$\omega^{s_A} \prod_{j\in A} (1 + (\omega X)^j) \prod_{j\in B} (1 + X^j) \quad \text{and} \quad \omega^{s_B} \prod_{j\in B} (1 + (\omega X)^j) \prod_{j\in A} (1 + X^j).$$

Each polynomial has degree W. The element w is a primitive n-th root and the order of four modulo n is h, so its minimal polynomial over $\mathbb{F}_4$ is $X^h + \omega^2$. Thus a collision with $W < h$ would make the two polynomials identical. Their constant coefficients first force $s_A \equiv s_B \pmod 3$. Cancel that common scalar and let j_0 be the least index in $A \cup B$. If, say, $j_0 \in A$, then the coefficients of X^{j_0} on the two sides are respectively ω^{j_0} and 1. They differ because $3 \nmid j_0$; the other case is symmetric. More sharply, if $s_A \not\equiv s_B \pmod 3$, the difference has span W, while if $s_A \equiv s_B \pmod 3$, its constant and leading terms cancel but its coefficients at degrees j_0 and $W - j_0$ are nonzero, so its span is $W - 2j_0$. A nonzero multiple of $X^h + \omega^2$ has span at least h. This proves both pairwise criteria in the statement.

When $s_m < h$, necessarily $m < n$, so $\mathcal{A}_m \subseteq \mathcal{U}_h$. For $E, F \subseteq \mathcal{A}_m$, their symmetric difference has weighted sum at most s_m. Hence all 2^{t_m} displayed powers are distinct.

Every subset E with $s_E \le R_h$ belongs to a family whose pairwise weighted symmetric differences have sum at most $2R_h < h$. The number of such subsets is exactly $\mathcal{P}_h$, proving the first finite bound. For its asymptotic, put

$$G(t) = \prod_{j \ge 1} (1 + e^{-tj}), \qquad F(e^{-t}) = \prod_{3 \nmid j} (1 + e^{-tj}) = \frac{G(t)}{G(3t)}.$$

The standard q-Pochhammer asymptotic gives

$$G(t) \sim 2^{-1/2} \exp\left(\frac{\pi^2}{12t}\right), \qquad \frac{F(e^{-t})}{1 - e^{-t}} \sim t^{-1} \exp\left(\frac{\pi^2}{18t}\right).$$

The coefficients of $F(X)/(1 - X)$ are the nonnegative, nondecreasing cumulative counts in $\mathcal{P}_h$. Ingham's exponential Tauberian theorem [7] therefore gives

$$\log \mathcal{P}_h = 2\sqrt{\frac{\pi^2 R_h}{18}} + o(\sqrt{h}) = \frac{\pi}{3}\sqrt{h} + o(\sqrt{h}).$$

Finally, $t_m = 2m/3 + O(1)$ and $s_m = m^2/3 + O(m)$; taking the largest m with $s_m < h$ gives the elementary initial-segment estimate. The final assertion follows because a missing full ℓ-part would force $\operatorname{ord}(\eta_k) \mid N/\ell$. □

There is a second, sharp consequence of the same selected formula. It classifies the first possible one-sided relation rather than merely packing many pairwise-distinct powers.

Proposition 6.5 (sharp selected first-relation threshold)**.** *Retain the notation above. For every nonempty $E \subseteq \mathcal{U}_h$, put $S = s_E$ and decompose*

$$B_E(X) = \prod_{j \in E} (1 + X^j) = B_0(X^3) + X B_1(X^3) + X^2 B_2(X^3), \qquad B_a \in \mathbb{F}_2[Y].$$

If $a_0 \equiv -S \pmod 3$, then the exact all-weight criterion is

$$\eta_k^{Q_E} = 1 \quad \Longleftrightarrow \quad Y^{2H} + Y^H + 1 \mid B_a(Y) \quad \textit{for both } a \ne a_0, \qquad H = h/3.$$

In particular,

$$\eta_k^{Q_E} = 1 \quad \Longrightarrow \quad s_E \ge 2h + 1.$$

The threshold is sharp: with $H = h/3$,

$$E_0 = \{1, h-1, h+1\}, \qquad Q_{E_0} = 1 + 2^H + 2^{2H} = |U_k|, \qquad s_{E_0} = 2h + 1.$$

Consequently, if a prime $\ell \mid N = |U_k|$ is absent from the selected order at its full valuation, write

$$\frac{N}{\ell} = \sum_{0 \le z < 2H} \epsilon_z 2^z, \qquad j_z \equiv 2^z \pmod{3h}, \quad 1 \le j_z < 3h.$$

Then necessarily

$$\sum_{\epsilon_z = 1} j_z \ge 2h + 1.$$

Proof. Put

$$P_E(X) = \omega^S B_E(\omega X) + B_E(X).$$

The selected formula shows that $\eta_k^{Q_E} = 1$ exactly when $M(X) = X^h + \omega^2$, the minimal polynomial of w over $\mathbb{F}_4$, divides P_E. If $[X^r]B_E = b_r \in \mathbb{F}_2$, then

$$[X^r]P_E = b_r(1 + \omega^{S+r}).$$

Thus the scalar multiplier vanishes precisely on the section $r \equiv -S \pmod 3$. Write the three sections as in the statement. Since $h = 3H$, division by $M(X) = (X^3)^H + \omega^2$ respects these three sections. Divisibility by M therefore forces each of the two surviving $B_a(Y)$ to be divisible by $Y^H + \omega^2$. Because its coefficients lie in $\mathbb{F}_2$, conjugation also forces divisibility by $Y^H + \omega$, and hence by

$$(Y^H + \omega^2)(Y^H + \omega) = Y^{2H} + Y^H + 1.$$

The converse is immediate because the displayed polynomial contains $Y^H + \omega^2$ as a factor. This proves the exact all-weight criterion.

If $S < 2h$, every surviving section has degree below $2H$. If $S = 2h$, the top term lies in the vanished residue-zero section, so the same strict bound holds for the two surviving sections. Hence both are zero. The constant term of B_E then forces the sole supported section to be residue zero, but the least $j \in E$ contributes coefficient one at X^j, in a nonzero residue class. This contradiction proves $S \ge 2h + 1$.

The congruence $2^H \equiv h - 1 \pmod{3h}$ follows by induction from $2 \equiv 2 \pmod 9$ after cubing; squaring gives $2^{2H} \equiv h + 1 \pmod{3h}$. Hence E_0 has the asserted exponent, and $\eta_k^N = 1$ proves sharpness. Finally, N is odd and prime to three, so $\ell \ge 5$ and $N/\ell < 2^{2H}$. Its binary indices lie below $2H$ and have distinct residues modulo $3h$. A missing full ℓ-part gives $\eta_k^{N/\ell} = 1$, so the first assertion gives the displayed residue-weight bound. $\square$

Together, the partition and first-relation bounds strengthen the polynomial Singer sieve to an exponential-in-$\sqrt{h}$ exclusion and a sharp weight-$2h$ exclusion for the actual norm-one coordinate. They remain subexponential and do not classify relations above the norm triple, so they do not force primitivity.

There is an exact number-field factorization of the remaining index. Put $L = \mathbb{Q}(\zeta_{3^{k+1}})^+$ and $K = \mathbb{Q}(\zeta_{3^k})^+$, reduce global units E_F and circular units C_F modulo the inert prime over two, and write

$$R_F = (\mathcal{O}_F/2)^\times/\overline{E}_F, \qquad T_F = \overline{E}_F/\overline{C}_F.$$

Then

$$[U_k : \langle \eta_k \rangle] = |\ker(\mathrm{N} : R_L \to R_K)| \frac{[\overline{E}_L : \overline{C}_L]}{[\overline{E}_K : \overline{C}_K]}.$$

The first factor is a relative principal 2-ray kernel; the second divides the relative real class-number quotient. Even class number one would leave the principal-ray factor. This is not a formal possibility only: analogous prime-conductor periods can have inert two and class number one while their finite circular-unit reduction has nontrivial defect [6]. Arm C is exactly the assertion that both factors in the displayed index formula vanish at every level.

7 The exceptional arm: corrected norm and Capelli

Let $h = 3^k$, $q = 2^h$, $\zeta = \zeta_k$, and $\omega = \zeta^h \in \mathbb{F}_4$.

Proposition 7.1 (first possible translate). *If $f(p) = 2 \cdot 3^k$, then $m_p \geq 4$.*

Proof. Such a prime divides $2^h + 1$ but neither 3^{k+1} nor $2^h - 1$. The half-angle decomposition in Proposition 6.1 shows that each of $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$ has order prime to p. All four are p-th powers in their defining fields, so the excess definition gives the claim. $\square$

The translate by four lies in a compositum with $\mathbb{F}_{16}$. The nontrivial relative Frobenius over $\mathbb{F}_4$ exchanges the nimbers 4 and 5, which are the roots of $Y^2 + Y + \omega$.

Proposition 7.2 (corrected norm). *In $\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}$,*

$$\mathrm{N}(\zeta + 4) = (\zeta + 4)(\zeta + 5) = \zeta^2 + \zeta + \omega = A_k.$$

Proof. One has $4 + 5 = 1$ and $4 \cdot 5 = \omega$. Expanding the product gives the formula. $\square$

Define

$$R_k(Y) = \mathrm{Res}_X \left(X^{2h} + X^h + 1,\ Y + X^h + X^2 + X \right).$$

Theorem 7.3 (exact exceptional target). *The polynomial R_k is the irreducible degree-$2h$ minimal polynomial of A_k. The prime divisors of $\Psi_k = \Phi_{2\cdot 3^k}(2)/3$ are exactly the primes $\ell \neq 3$ with $\mathrm{ord}_\ell(2) = 2h$, with their full valuations. The following are equivalent:*

1. *every row with $f(\ell) = 2 \cdot 3^k$ has $m_\ell = 4$;*
2. *$\Psi_k \mid \mathrm{ord}(M_k)$;*
3. *A_k is not an ℓ-th power in $\mathbb{F}_{2^{2h}}$ for every prime $\ell \mid \Psi_k$;*
4. *$R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.*

Proof. The element ζ has minimal polynomial $X^{2h} + X^h + 1$. To justify the degree of A_k, suppose $A_k^{2^j} = A_k$, put $t \equiv 2^j \pmod{3h}$, and write $u = \zeta^t$. If j is even, then $\omega^t = \omega$ and

$$(u + \zeta)(u + \zeta + 1) = 0,$$

so u is either ζ or $\zeta + 1$. If j is odd, then $\omega^t = \omega^2$ and

$$(u + \zeta + \omega)(u + \zeta + \omega^2) = 0,$$

so u is either $\zeta + \omega$ or $\zeta + \omega^2$. The three translated possibilities are impossible: Proposition 6.1 gives each the same order as $\zeta + 1$, namely $3^{k+1}\,\mathrm{ord}(\gamma_k) > 3^{k+1}$; here $\mathrm{ord}(\gamma_k) > 1$ because $\mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}$ and $h > 1$. On the other hand, u is a 3^{k+1}-st root of unity. Hence $u = \zeta$, so $2^j \equiv 1 \pmod{3h}$. Since the order of two modulo $3h = 3^{k+1}$ is $2h$, the Frobenius orbit of A_k has size $2h$. Thus A_k has degree $2h$ and the displayed resultant is its minimal polynomial.

The cyclotomic divisor criterion applied to $\Phi_{2\cdot 3^k}(2)$ gives the stated prime support after removing its single factor of three. Proposition 7.1 and the corrected norm give

$$m_\ell = 4 \quad\Longleftrightarrow\quad A_k \notin (\mathbb{F}_{2^{2h}}^{\times})^\ell.$$

Raising to $q-1$ is invertible on the current ℓ-Kummer quotient, so this is equivalent to the full ℓ-primary part occurring in $\operatorname{ord}(M_k)$. Finally, μ_ℓ lies in the defining field; Kummer theory and Capelli's lemma identify nonpower with irreducibility of $R_k(X^\ell)$. □

The exceptional Euler test also has a lossless sparse form parallel to the three-block criterion in the cubic arm.

Proposition 7.4 (sparse exceptional block criterion)**.** *Let $n = 3h$, let $\ell \mid \Psi_k$ be prime, put $d = (q+1)/\ell$, and work in $\mathcal{R}_h = \mathbb{F}_2[X]/(X^n - 1)$. With x the class of X, set*

$$\mathcal{A} = x^2 + x + x^h, \qquad \iota(x) = x^{-1}, \qquad C = \mathcal{A}^d.$$

Write the unique cyclic representative of $V = C + \iota(C)$ as

$$V = V_0 + X^h V_1 + X^{2h} V_2, \qquad \deg V_i < h.$$

Then

$$M_k^{(q+1)/\ell} = 1 \quad\Longleftrightarrow\quad V_0 = V_1 = V_2.$$

If $d = \sum_{i=0}^{h-1} \epsilon_i 2^i$, then, losslessly in $\mathcal{R}_h$,

$$C = \prod_{\epsilon_i = 1} \left(x^{2^{i+1}} + x^{2^i} + x^{h2^i} \right),$$

where exponents are read modulo n. Thus arm D is equivalent to failure of this three-block equality for every current prime ℓ.

Proof. The order of two modulo $3^{k+1} = n$ is $2h$, so $q = 2^h \equiv -1 \pmod{n}$. Hence q-powering on $\mathcal{R}_h$ is ι. In the selected field $\mathbb{F}_2[X]/(J_h)$, where $J_h = X^{2h} + X^h + 1$, the image of $\mathcal{A}$ is A_k, and

$$M_k^d = 1 \quad\Longleftrightarrow\quad (A_k^d)^q = A_k^d \quad\Longleftrightarrow\quad J_h \mid C + \iota(C).$$

The first equivalence is kernel-checked abstractly as `exceptional_euler_fixed_iff`. Modulo J_h, characteristic two gives $X^{2h} = X^h + 1$, and therefore

$$V \equiv (V_0 + V_2) + X^h (V_1 + V_2).$$

The right side has degree less than $2h = \deg J_h$, so it vanishes exactly when $V_0 = V_1 = V_2$. The product formula is the binary-power expansion of $\mathcal{A}^d$ in characteristic two. □

Corollary 7.5 (primitive-only palindromicity)**.** *In the notation of Proposition 7.4, one always has*

$$V_0 + V_1 + V_2 = 0.$$

Consequently the following conditions are equivalent:

$$M_k^d = 1, \qquad V_0 = V_1 = V_2, \qquad V_0 = V_1 = V_2 = 0, \qquad C = \iota(C) \text{ in } \mathcal{R}_h.$$

In particular, every proper cyclic fold through $\mathbb{F}_2[X]/(X^h - 1)$ vanishes identically on V; no Fourier test on a proper 3-power quotient can detect arm D.

Proof. By lifting the exponent of three,

$$v_3(2^h+1)=k+1,$$

so $n=3^{k+1}$ divides $q+1$. Since $\ell\neq 3$, it follows from $q+1=\ell d$ that $n\mid d$. Modulo X^h-1 one has

$$\mathcal{A}=X^2+X+1, \qquad \iota(\mathcal{A})=X^{-2}\mathcal{A}.$$

Hence

$$\iota(C)=X^{-2d}C=C \pmod{X^h-1}.$$

The image of $V_0+X^hV_1+X^{2h}V_2$ in this quotient is $V_0+V_1+V_2$, proving the first assertion. If the three blocks are equal to W, their sum is $3W=W$ in characteristic two, so $W=0$. The converse is immediate, and every quotient X^m-1 with $m\mid h$ factors through X^h-1. $\square$

There is a second, complementary compression which shortens the binary product by one third before testing this palindromicity.

Proposition 7.6 (one-third Frobenius compression)**.** *Put*

$$H=3^{k-1}, \qquad h=3H, \qquad Q=2^H, \qquad S=Q^2-Q+1=3\Psi_k,$$

and, for a prime $\ell\mid\Psi_k$, put $e=S/\ell$. Then

$$Q\equiv h-1 \pmod{3h}, \qquad S\equiv 3 \pmod{3h}, \qquad e\equiv 3 \pmod{3h},$$

and

$$d=(Q+1)e.$$

In $\mathcal{R}_h$, define

$$\widetilde{\mathcal{B}}=\mathcal{A}^{Q+1}.$$

It has the lossless seven-term representative

$$\widetilde{\mathcal{B}}=1+x^{-2}+x^h+x^{h+1}+x^{2h}+x^{2h+1}+x^{2h+2},$$

and $C=\widetilde{\mathcal{B}}^e$. In the primitive component $\mathbb{F}_2[X]/(J_h)$ this reduces further to the trinomial

$$\mathcal{B}=x^{-2}+x+x^{2h+2}.$$

Thus

$$M_k^d=1 \iff \mathcal{B}^e=\iota(\mathcal{B})^e \text{ in } \mathbb{F}_2[X]/(J_h),$$

while arm D asks for the opposite inequality. The exponent now has at most $2H+1=2h/3+1$ binary places, rather than $h+1$.

If $e=3$—equivalently, if $\Psi_k=\ell$ is prime—then the opposite inequality always holds. Hence every prime-complement level satisfies arm D; in particular this gives proof-level, non-enumerative certificates for $k=2$ and $k=3$.

More generally, put $r=\Psi_k/\ell$ and

$$\varepsilon_k=\begin{cases}1, & k \text{ even},\\ \omega^2, & k \text{ odd}.\end{cases}$$

For the reciprocal ratio

$$P = \frac{\iota(\mathcal{B})}{\mathcal{B}} = M_k^{Q+1}$$

one has the exact identities

$$P^{\Psi_k} = \varepsilon_k, \qquad M_k^d = 1 \quad \Longleftrightarrow \quad P^r = \varepsilon_k.$$

Thus, after removing its explicitly known three-primary phase, the unresolved composite target has exponent $r = \Psi_k/\ell \equiv 1 \pmod{2h}$ *rather than* $e = 3\Psi_k/\ell$.

Proof. The congruence for Q follows by induction. The base case is $2 \equiv 3-1 \pmod 9$. If $Q+1 = h(1+3t)$, write $h = 3s$ and cube. Then

$$Q^3 + 1 = 3h\left(1 + 3\left[t - s(1+3t)^2 + s^2(1+3t)^3\right]\right),$$

which is the required congruence at the next level. Substitution gives

$$S \equiv (h-1)^2 - (h-1) + 1 \equiv 3 \pmod{3h}.$$

Since $S = 3\Psi_k$ and $\ell \mid \Psi_k$, the integer e is divisible by three. Moreover $\mathrm{ord}_\ell(2) = 2h$ implies $h \mid \ell - 1$. Therefore $3h \mid (\ell-1)e$, and from $\ell e = S$ one gets $e \equiv S \equiv 3 \pmod{3h}$.

The factorization $Q^3 + 1 = (Q+1)S$ gives $d = (Q+1)e$. Frobenius and the first congruence give

$$\mathcal{A}^Q = x^{2h-2} + x^{h-1} + x^{2h}.$$

Multiplication by $\mathcal{A} = x^2 + x + x^h$ cancels the two copies of x^{2h-1} and gives the displayed seven terms. Modulo J_h, the three terms $1 + x^h + x^{2h}$ vanish and $x^{h+1} + x^{2h+1} = x$, leaving $\mathcal{B}$. The fixed-field equivalence is then Proposition 7.4 and Corollary 7.5 applied to $C = \widetilde{\mathcal{B}}^e$.

We next determine the three-primary phase. Since $(Q+1)\Psi_k = (q+1)/3$,

$$P^{\Psi_k} = M_k^{(q+1)/3} = A_k^{(q^2-1)/3} = \mathrm{N}_{\mathbb{F}_{4^h}/\mathbb{F}_4}(A_k).$$

The minimal polynomial of ζ over $\mathbb{F}_4$ is $X^h + \omega$. If $y^2 + y + \omega = 0$, its other conjugate is $y+1$, so the resultant formula for the norm gives

$$\mathrm{N}(A_k) = (y^h + \omega)((y+1)^h + \omega).$$

Reduction by $y^2 + y + \omega = 0$ gives $y^{15} = 1$ and

h mod 15	3	9	12	6
y^h	$\omega + \omega^2 y$	$\omega + \omega y$	$1 + \omega^2 y$	$\omega y.$

The residues of 3^k modulo 15 run through $3, 9, 12, 6$. Using $y(y+1) = \omega$, the displayed product is therefore ω^2 in the first and third cases and 1 in the second and fourth. This proves $P^{\Psi_k} = \varepsilon_k$.

Now $e = 3r$. If $P^e = 1$, then P^r is a cube root of unity. Moreover

$$(P^r)^\ell = P^{\Psi_k} = \varepsilon_k.$$

Every current prime satisfies $\ell \equiv 1 \pmod 3$, so raising a cube root of unity to the ℓ-th power fixes it. Hence $P^r = \varepsilon_k$. The converse follows from $\varepsilon_k^3 = 1$. Finally, since every prime factor of r is current and hence congruent to one modulo $2h$, one has $r \equiv 1 \pmod{2h}$.

It remains to prove the prime-complement assertion, where $r = 1$. Put $z = \zeta$ and multiply numerator and denominator of P by z^2. Then

$$P = \frac{N}{D}, \qquad N = z^4 + z + \omega, \qquad D = 1 + z^3 + \omega^2 z^4,$$

and direct factorization over $\mathbb{F}_4$ gives

$$N + D = \omega(z + \omega^2)(z^3 + \omega^2), \qquad N + \omega^2 D = \omega^2(z + 1)(z^3 + \omega).$$

If k is even, the first identity shows $P \neq 1$; if k is odd, the second shows $P \neq \omega^2$. Indeed z has order $3h \geq 27$, so neither z nor z^3 can equal the displayed cube root of unity. Thus $P \neq \varepsilon_k$, as required. The only earlier level has $\Psi_1 = 1$ and no current prime. $\square$

The parity-twisted exponent reduction does not by itself settle composite Ψ_k. After putting $U = \varepsilon_k^{-1} P$, it says exactly that $U^{\Psi_k} = 1$ and that arm D asks for $U^{\Psi_k/\ell} \neq 1$ for every current prime ℓ. Thus it isolates the prime-to-three current coordinate but does not prove that every one of its prime factors occurs in $\operatorname{ord}(U)$. Here the normalization uses $\Psi_k \equiv 1 \pmod 3$ and $r \equiv 1 \pmod 3$, the latter already following from $r \equiv 1 \pmod{2h}$.

The ordinary-arm ratio packing has an exceptional analogue, provided the multiplier roots of unity and the degree of the marked root are balanced by an intermediate constant field.

Proposition 7.7 (intermediate-field exceptional packing sieve). *Assume $k \geq 2$, put $h = 3^k$, and let the prime ℓ divide Ψ_k. For $0 \leq j < k$, set*

$$m_j = 3^j, \qquad R_j = 3^{k-j}, \qquad s_j = \left\lfloor \frac{R_j - 1}{8} \right\rfloor$$

and define

$$\mathcal{S}_k = \max_{0 \leq j < k} \sum_{t=0}^{\min(2m_j, s_j)} 2^t \binom{2m_j}{t} \binom{s_j}{t}.$$

If the current ℓ-coordinate is deficient, then

$$\frac{\Psi_k}{\ell} \geq \mathcal{S}_k.$$

Consequently arm D holds at every current factor satisfying

$$\frac{\Psi_k}{\ell} < \mathcal{S}_k.$$

Moreover, if $H_2(x) = -x \log_2 x - (1 - x) \log_2(1 - x)$ is binary entropy, then

$$\log_2 \mathcal{S}_k \geq c_0 \sqrt{h} - O(\log h), \qquad c_0 = \frac{1}{4} + \frac{2}{3} H_2\left(\frac{3}{8}\right) + \frac{3}{8} H_2\left(\frac{2}{3}\right) > 1.2306.$$

Thus this sieve proves an exponential-in-$\sqrt{h}$ large-factor range at every exceptional level, while remaining subexponential in h.

Proof. Write $z = \zeta$, $d = \Psi_k/\ell$, and regard the normalized phase as the rational function

$$U(T) = \varepsilon_k^{-1} \frac{\mathcal{N}(T)}{\mathcal{D}(T)}, \qquad \mathcal{N}(T) = T^4 + T + \omega, \qquad \mathcal{D}(T) = 1 + T^3 + \omega^2 T^4.$$

The polynomial $T^h + \omega$ is the minimal polynomial of z over $\mathbb{F}_4$. The order of four modulo $3h$ is h, and the map

$$t \longmapsto z^{4^t-1}, \qquad 0 \le t < h,$$

is a bijection from the Frobenius orbit indices to μ_h: every value is in μ_h, and injectivity follows from the orbit size. Hence the roots of $T^h + \omega$ are exactly cz, $c \in \mu_h$. Since U is defined over $\mathbb{F}_4$, deficiency $U(z)^d = 1$ implies

$$U(cz)^d = 1 \qquad (c \in \mu_h).$$

Fix j, abbreviate $m = m_j$, $R = R_j$, $s = s_j$, and put $E = \mathbb{F}_{4^m}$. LTE gives

$$v_3(4^m - 1) = 1 + v_3(m) = j + 1,$$

so $\mu_{3m} \subset E$. Since $E \subset \mathbb{F}_{4^h}$, $[E(z) : E] = h/m = R$. Also z^R has order $3m$ and lies in E, so the minimal polynomial of z over E is the binomial $T^R + z^R$.

Over $\mathbb{F}_4$ introduce the six monic quadratics

$$\begin{aligned} &f_1 = T^2 + \omega T + 1, \quad f_2 = T^2 + \omega T + \omega, \quad f_3 = T^2 + \omega^2 T + 1, \\ &f_4 = T^2 + T + \omega, \qquad f_5 = T^2 + T + \omega^2, \quad\; f_6 = T^2 + \omega^2 T + \omega^2. \end{aligned}$$

They are distinct and irreducible by direct inspection in $\mathbb{F}_4$. Up to nonzero scalar factors,

$$\mathcal{N}(T) = f_1(T)f_2(T), \qquad \mathcal{D}(T) \doteq f_3(T)f_4(T),$$

and

$$\mathcal{N}(\omega T) \doteq f_4(T)f_5(T), \qquad \mathcal{D}(\omega T) = f_2(T)f_6(T),$$

where $\doteq$ denotes equality up to a nonzero scalar. Since m is odd, $E \cap \mathbb{F}_{16} = \mathbb{F}_4$, so all six quadratics remain irreducible over E.

Choose representatives $\mathcal{C}$ for μ_{3m}/μ_3. Scaled factors belonging to different representatives are disjoint. Indeed, a common root of $f_a(cT)$ and $f_b(c'T)$ would give $c'/c \in \mathbb{F}_{16}^\times \cap \mu_{3m} = \mu_3$, hence $c = c'$ by the choice of representatives, and then $a = b$ because the six factors are distinct. In particular, for each $c \in \mathcal{C}$ the two elements

$$u_{c,0} = U(cz), \qquad u_{c,1} = U(\omega cz)$$

have, up to scalars, the factor patterns

$$u_{c,0} \doteq \frac{f_1(cz)f_2(cz)}{f_3(cz)f_4(cz)}, \qquad u_{c,1} \doteq \frac{f_4(cz)f_5(cz)}{f_2(cz)f_6(cz)}.$$

Their denominators are nonzero: otherwise a conjugate of z would be a root over $\mathbb{F}_4$ of the degree-four polynomial $\mathcal{D}$, whereas z and all its conjugates have degree $h \ge 9$. All $2m$ displayed elements are d-th roots of unity.

Index them by a set I of size $2m$. For $v \in \mathbb{Z}^I$ with $\sum_i |v_i| \le s$, put $V_v = \prod_i u_i^{v_i}$. Suppose $V_v = V_w$ and set $g = v - w$. Separating positive and negative exponents and cross-multiplying gives an equality at z between two polynomials in $E[T]$, each of degree at most

$$4 \sum_i |g_i| \le 8s < R.$$

The degree-R minimal polynomial therefore forces equality of the two polynomials. Unique factorization now recovers every exponent: for a fixed c, the valuation at $f_1(cT)$ gives $g_{c,0} = 0$, and the valuation at $f_5(cT)$ gives $g_{c,1} = 0$. Cross-star disjointness handles all $c \in \mathcal{C}$. Thus the V_v are distinct.

The signed ℓ^1 ball in $\mathbb{Z}^{2m}$ has exactly

$$\sum_{t=0}^{\min(2m,s)} 2^t \binom{2m}{t}\binom{s}{t}$$

points. Since their images are distinct roots of $X^d - 1$, this number is at most d. Taking the maximum over j proves the exact bound.

For the asymptotic statement, first let k be even and take $j = k/2 - 1$. Then, with $x = \sqrt{h}$,

$$2m = \frac{2}{3}x, \qquad s = \frac{3}{8}x + O(1).$$

The summand with $t = \lfloor x/4 \rfloor$ and Stirling's formula give

$$\log_2 \mathcal{S}_k \geq \left[\frac{1}{4} + \frac{2}{3}H_2\left(\frac{3}{8}\right) + \frac{3}{8}H_2\left(\frac{2}{3}\right)\right] x - O(\log h).$$

If k is odd, take $j = (k-3)/2$. Then

$$2m = \frac{2}{3\sqrt{3}}x, \qquad s = \frac{3\sqrt{3}}{8}x + O(1),$$

and the same choice $t = \lfloor x/4 \rfloor$ gives coefficient

$$\frac{1}{4} + \frac{2}{3\sqrt{3}}H_2\left(\frac{3\sqrt{3}}{8}\right) + \frac{3\sqrt{3}}{8}H_2\left(\frac{2}{3\sqrt{3}}\right) > 1.2341 > c_0.$$

The $O(\log h)$ errors are the standard logarithmic errors in the two binomial-coefficient estimates. This proves the uniform claim for all sufficiently large k; enlarging the implicit constant absorbs the finitely many smaller levels. $\square$

This packing theorem is a genuine composite-Ψ_k range, not a universal proof: its lower bound is exponential in $\sqrt{h}$, whereas Ψ_k is exponential in h. Logarithmic or Hasse derivatives do not add an intrinsic constraint in this finite etale quotient: $\mathbb{F}_2[X]/(J_h)$ has no nonzero $\mathbb{F}_2$-derivations, while the formal polynomial derivative does not descend because $J_h' = X^{h-1}$ is a unit modulo J_h. A derivative argument would therefore need genuinely new lift or multiplicity information. Proposition 7.6 and Proposition 7.7 therefore prove the prime-complement case and a large-factor range, not universal arm D.

Proposition 7.8 (effective exceptional bound)**.** *Let $k \geq 2$, $h = 3^k$, and let $\ell \mid \Psi_k$ be prime. If t is the least power of two satisfying $t \geq \lceil \log_2(h-1) \rceil$, then*

$$m_\ell \leq 4^t - 1 < 16h^4.$$

Proof. Put $K = \mathbb{F}_{4^h}$ and extend constants to $L = K\mathbb{F}_{4^t}$. For a character χ of exact order ℓ, the sum

$$S = \sum_{c \in \mathbb{F}_{4^t}} (\chi \circ \mathrm{N}_{L/K})(\zeta + c)$$

equals 4^t precisely when every translate is an ℓ-th power. The associated primitive even function-field character has L-polynomial $(1-T)\prod_{j=1}^{h-2}(1-\alpha_j T)$ with $|\alpha_j| = 2$. Constant-field extension therefore gives

$$|S| \le 1 + (h-2)2^t < 4^t$$

when $2^t \ge h-1$; see [14]. Some translate is a nonpower, and norm injectivity on the ℓ-Kummer quotient returns the statement to its defining field. The elements of $\mathbb{F}_{4^t}$ are the finite nimbers below 4^t. The displayed polynomial estimate follows from the minimal choice of t. □

This bound is level-dependent and does not prove absolute boundedness or the value four.

The exceptional coordinate also has an exact circular-ray factorization. In the real field generated by roots of orders 3^{k+1} and 5, let $\mathfrak{P}$ be the prime selected by $\zeta \mapsto \zeta_k$ and the conductor-five coordinate $\mapsto \omega$. Let E_L and C_L be the global and circular units and let a subscript ℓ denote the primary projection after reduction. Then

$$\overline{C}_{L,\ell} = \langle A_{k,\ell}\rangle = \langle M_{k,\ell}\rangle,$$

and

$$[k_{\mathfrak{P},\ell} : \langle M_{k,\ell}\rangle] = \left|\ker\bigl(\mathrm{Cl}_L(\mathfrak{P})_\ell \to \mathrm{Cl}_{L\ \ell}\bigr)\right| \, |(\overline{E}_L/\overline{C}_L)_\ell|.$$

The second factor is controlled by the real class number, but the first is a principal ray kernel. The marked conductor-five cyclotomic unit is globally not an ℓ-th power; the height theorem of Amoroso–Dvornicich gives this directly [1]. Global nonpower does not determine its Frobenius at the selected prime. Arm D is exactly the assertion that both factors in the displayed ray-class index vanish.

The current C and D coordinates are orthogonal. If $q_0 = 2^{h/3}$ and $q = q_0^3$, their norm-one tori have orders

$$q_0^2 + q_0 + 1 \quad \text{and} \quad q+1,$$

which are coprime. The exceptional cubical operator kills the cubic subfield, while the local norm kills the exceptional torus. Hence the two principal-ray factors cannot force one another through the shared prime over two. Each requires its own marked unit-image calculation.

Even a perfect one-dimensional Kummer–Artin pairing does not show that a prescribed prime has nonzero image in the Artin quotient. The obstruction already occurs arithmetically.

Proposition 7.9 (perfect-pairing no-go)**.** *There exist a number field K containing μ_3, a global unit u which is not a cube in K, and exactly one prime $\mathfrak{q}$ of K above two; this prime splits completely in the nontrivial cyclic Kummer extension $K(\sqrt[3]{u})/K$.*

Proof. Take

$$K = \mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\zeta_3, \sqrt{3}), \qquad u = 2 + \sqrt{3}.$$

The Pell equation gives $\mathcal{O}^\times_{\mathbb{Q}(\sqrt{3})} = \{\pm u^n : n \in \mathbb{Z}\}$, so u is not a cube in the real quadratic subfield. If $X^3 - u$ acquired a root in its quadratic extension K, its cubic factorization over the real subfield would also have a linear factor. Hence u is not a cube in K, and adjoining a cube root gives a nontrivial cyclic cubic extension.

Locally at two, $\mathbb{Q}_2(\zeta_3)/\mathbb{Q}_2$ is unramified quadratic and adjoining i is ramified quadratic. The two local quadratics are linearly disjoint, so $K \otimes \mathbb{Q}_2$ is a degree-four field; hence two has a unique prime $\mathfrak{q}$ in K. In its residue field $\sqrt{3} \equiv 1$, so $u \equiv 1 \pmod{\mathfrak{q}}$. The polynomial $X^3 - u$ has the simple root 1 modulo $\mathfrak{q}$, because its derivative at one is the unit 3. Hensel's lemma makes u a cube in $K_{\mathfrak{q}}$; since $\mu_3 \subset K_{\mathfrak{q}}$, the polynomial splits completely there and $\mathfrak{q}$ splits completely in $K(\sqrt[3]{u})$. □

Thus nontriviality of the Kummer line, one-dimensionality of its dual, and even uniqueness of the prime above two are compatible with zero marked pairing. In the four arms, perfect duality reduces the target to nonvanishing of the Conway-selected prime class; it does not prove that nonvanishing.

8 Nim reciprocity saturation and marked holonomy

The residue-symbol reductions suggest using reciprocity to evaluate the remaining phase. There is a precise saturation boundary for that program. Let ℓ be an odd prime, let $s \in \mu_\ell$, and put $E = \mathrm{ord}_\ell(2)$. The two-axis Frobenius–cyclotomic family is $s_{j,c} = s^{2^j c}$ for $0 \le j < E$ and $c \in \mathbb{F}_\ell^\times$.

Proposition 8.1 (reciprocity saturation)**.** *The following statements hold.*

1. *Every multiplicative word in the $s_{j,c}$ is s^λ, where*

$$\lambda = \sum_{j,c} w_{j,c} 2^j c \pmod{\ell}.$$

   It is blind when $\lambda = 0$ and, when $\lambda \neq 0$, equals one exactly when the original marked phase equals one.

2. *For multiplicative characters χ, ψ of a finite field and $a \neq 0$, the Jacobi convolution*

$$J_{\chi,\psi}(a) = \sum_x \chi(x)\psi(a - x)$$

   satisfies

$$J_{\chi,\psi}(a) = \chi(a)\psi(a) J_{\chi,\psi}(1).$$

   In particular, if both characters are powers of one order-ℓ character, the marked dependence is again one power of the same phase.

3. *In the affine group on $\mathbb{F}_\ell$, write (t, q), with $q \in \mathbb{F}_\ell^\times$, for the map $x \mapsto qx + t$. If $q \neq 1$, all (t, q) are conjugate. Consequently every class function is independent of t.*

Proof. The first statement follows by collecting exponents. For the second, substitute $x = ay$ and factor $\chi(a)\psi(a)$ from every summand. For the third, conjugation by the translation $x \mapsto x + a$ changes t to $t + (1 - q)a$; taking $a = t/(q - 1)$ makes this zero. □

Thus norm products, weighted Stickelberger words, and Gauss or Jacobi transforms either erase or re-encode one phase. A class function on the single fixed-multiplier affine coset is blinder still: it forgets the translation coordinate in part 3. None of these operations independently evaluates the selected phase.

Ordinary cross-arm multiplicative words face the same boundary. If a strict ancestor lies in a field $\mathbb{F}_{2^d}$ with $f(\ell) \nmid d$, exactness of the residue degree gives $\ell \nmid 2^d - 1$; the ℓ-power map is therefore an automorphism on that ancestor group and its ℓ-Kummer image is trivial. Thus every cross-arm word built from strict lower-level coordinates in such degrees separates its coprime-primary torsion. Nor does the next ℓ-component give an induction: its defining radicand is $\alpha_\ell = \kappa_{f(\ell)} + m_\ell$, so irreducibility proves only that the actual winning translate is a non-ℓ-th power. Using that fact to identify m_ℓ with the conjectural 0, 1, or 4 would already assume the target.

The intrinsic coordinate is a full Frobenius-orbit holonomy, not a one-step affine translation. Let $\beta \in \mathbb{F}_{2^E}^\times$ be the selected element, put $s = \beta^{(2^E-1)/\ell}$, choose $\alpha^\ell = \beta$ in an algebraic closure, fix a primitive $\zeta \in \mu_\ell$, and let $\varphi(z) = z^2$. Then

$$\frac{\varphi^E(\alpha)}{\alpha} = \alpha^{2^E-1} = \beta^{(2^E-1)/\ell} = s.$$

This ratio is unchanged when α is replaced by $\zeta^a\alpha$, because φ^E fixes μ_ℓ. If $s = \zeta^T$, the E-step action on the labeled roots is the pure translation $e_x \mapsto e_{x+T}$. Hence on the Fourier lines

$$v_r = \sum_x \zeta^{-rx} e_x, \qquad \varphi^E(v_r) = \zeta^{rT} v_r.$$

The diagonal eigenvalue at $r = 1$ is exactly the selected Kummer–Artin symbol.

Equivalently, choose a root frame over each conjugate $\beta_j = \beta^{2^j}$ and write the one-step maps as $e_{j,x} \mapsto e_{j+1,2x+t_j}$. Their cyclic holonomy is

$$T = \sum_{j=0}^{E-1} 2^{E-1-j} t_j \pmod{\ell}.$$

Changing frames by $\alpha_j \mapsto \zeta^{a_j}\alpha_j$ sends t_j to $t_j + 2a_j - a_{j+1}$; with $a_E = a_0$, these changes telescope because $2^E = 1$ modulo ℓ. Thus T is intrinsic, while the individual off-diagonal coefficients

$$\rho(t_j, 2)v_r = \zeta^{r2^{-1}t_j} v_{r2^{-1}}$$

are gauge-dependent. Every proper edge can be normalized away recursively, but a closed gauge cannot erase the cyclic holonomy: all but one edge may be made trivial, with the remaining edge carrying T.

This also explains the cocycle no-go. A homomorphic splitting on a commutative multiplier subgroup obeys $t(qr) = t(q) + qt(r)$. Fixing $q_0 \neq 1$ and comparing $t(qq_0)$ with $t(q_0q)$ gives

$$t(q) = (1-q)a, \qquad a = \frac{t(q_0)}{1-q_0}.$$

Every such cocycle is a coboundary and has trivial full-period transfer. Powers of one affine lift give this law on the subgroup generated by its multiplier. A general labeled section has a factor set, but changing root frames changes that factor set by a coboundary; only its cyclic holonomy is gauge-invariant.

Affine ancestry between two such torsor paths cannot create a missing phase. If the vertex maps are $x \mapsto \lambda x + b_j$, then the target translations have the form

$$u_j = \lambda t_j + b_{j+1} - 2b_j,$$

and the path holonomies satisfy

$$U_n = \lambda T_n + b_n - 2^n b_0.$$

On a closed E-cycle the endpoint term telescopes, so $U = \lambda T$. Consequently an affine recursion from a strict ancestor with trivial ℓ-holonomy cannot prove a nonzero current phase; a successful relation must contain a genuinely non-affine or independently nonzero closing source.

A successful nim-reciprocity proof must therefore retain and independently evaluate the marked holonomy T, equivalently s, along the literal Conway ancestry. Constructing the holonomy from the selected phase is an exact reformulation, not yet its evaluation.

9 Boundedness and the remaining theorem

Lenstra asked whether the finite excesses are absolutely bounded [8]. Conjecture 1.1 would give the bound four, but boundedness is strictly weaker than the maximal-order assertions above.

For $a \geq 0$, let $C_a = \mathbb{F}_{2^{2^a}}$. For an odd prime p, put $x_p = \kappa_{f(p)}$ and $L_{p,a} = \mathbb{F}_2(x_p)C_a$.

Proposition 9.1 (finite-line criterion)**.** *For every p and a,*

$$m_p < 2^{2^a} \quad \Longleftrightarrow \quad \textit{some } c \in C_a \textit{ has } x_p + c \notin L_{p,a}^p.$$

Consequently,

$$\sup_p m_p < \infty \quad \Longleftrightarrow \quad \textit{one fixed } a \textit{ works for every } p.$$

Proof. For $F_c = \mathbb{F}_2(x_p + c)$, one has $F_c C_a = L_{p,a}$ and $[L_{p,a} : F_c] \mid 2^a$, which is prime to p. Extension is therefore injective on p-Kummer quotients. The finite nimbers below 2^{2^a} are exactly C_a, proving the first equivalence from the excess definition. A fixed a gives a uniform bound; conversely choose a above any given uniform bound. □

No fixed finite affine line is known to work for every prime. The complete universal frontier remains the four selected order assertions in Theorem 3.1:

1. full primary order for the synchronized structural norm, including every Conway–Fermat equality $\delta_n = F_n$;
2. full primary projective order for the marked ordinary cyclotomic unit;
3. primitivity of every selected Gaussian period γ_k, equivalently generation by every η_k;
4. $\Psi_k \mid \mathrm{ord}(M_k)$ at every exceptional level.

Each is now one-dimensional and ancestry-sensitive. Degree, trace, norm, conductor, factor parity, or a statement about generic ambient points does not decide it.

10 Authority and verification

The claims in this paper have six distinct sources. Table 1 is part of the theorem statement: in particular, a finite calculation or a Lean definition of a target is never used as evidence for a universal claim.

The Lean 4 [4] development is exposed by the import-only entry point `Ogdoad.Papers.Excess` and uses Mathlib [15]. Its certificate namespaces share the single evaluator `Ogdoad.Certificate.BinaryPolynomial`; the moduli, field elements, and propositions remain local to the certificate they identify. The identifier `DPrimeTarget`, like the analogous selected targets, merely defines an open proposition. It is neither an axiom nor a theorem. The formalization therefore verifies ingredients of Theorem 3.1, not an end-to-end proof of Conjecture 1.1.

The maintained finite record is as follows.

- The 126 values for odd primes $3 \leq p \leq 709$ are source-backed by OEIS A380496 [12]; they are data, not an induction.
- The rows $m_{359} = m_{719} = m_{727} = 1$ have separate exact local finite-field certificates.

Status	Content
Cited input	Conway–Lenstra tower formulas, standard finite-field and Capelli theory, and the cited cyclotomic, class-field, function-field, and order results. These are not reproved in Lean.
Proved here	The four-arm equivalence; the exact selected coordinates and order criteria; the primitive-support, packet, supersingular, Singer and selected norm-one partition and first-relation bounds, corrected-norm, ordinary section and Kummer-ratio packing sieves, exceptional palindromicity, one-third compression, intermediate-field packing, prime-complement exceptional case, ray-index, boundedness, and reciprocity-saturation results stated above.
Ordinary kernel-checked Lean	Algebra without native evaluation: the algebraic Kummer interface, full-primary power tests, selected cubic and exceptional identities, low-degree minimal-polynomial collision, cubical reductions, packet and Fibonacci identities, and marked-holonomy calculus. These proofs may still use Lean's standard logical dependencies `propext`, `Classical.choice`, and `Quot.sound`.
Native evaluation	Named finite arithmetic propositions proved with `native_decide`. Lean's axiom report records the generated `_native.native_decide.ax_1_1` for these declarations, so they are exact replayable finite calculations, not ordinary kernel reductions.
External evidence	Source-backed rows, Python certificates, and explicit countermodels to overstrong intermediate lemmas. These establish only their stated finite instances.
Open	The universal assertions Z, O, C, D, Conjecture 1.1, and absolute boundedness of m_p.

Table 1: Claim authority for the finite-excess paper.

- In the cubic arm, C_1, C_2, C_3 have exact order proofs and C_4, C_5 have complete finite certificates; C_6 retains a source-assisted factorization boundary.
- In the exceptional arm, D'_2, D'_3 follow analytically and D'_4, D'_5 have complete finite certificates; D'_6 retains a source-assisted primality boundary.
- Two-block tensor arguments prove the multicomponent zero arm at $h = 12, 15, 24, 36, 40$. They are theorem-level finite cases, not a universal generation theorem.

An explicit countermodel in a non-Conway field can refute a proposed proof from trace, norm, conductor, or factor shape. It is not a counterexample to the 0/1/4 rule unless it occurs on the literal recursively selected Conway ancestry.

11 Conclusion

The finite-excess problem is now four exact selected-order problems. The zero arm needs a synchronized primitive-support coordinate, the ordinary arm one Artin symbol on the Conway-selected inert ray, and the cubic and exceptional arms the faithful finite reductions of two marked units. Every reduction retains the full primary valuation; none replaces the selected coordinate by a generic element of the same field.

This is also the obstruction to the presently available proof routes. Trace, norm, conductor, class group, and reciprocity packages can reconstruct or transport the marked phase, but the saturation theorem shows that homogeneous transforms of that same phase do not evaluate it. A proof of the 0/1/4 rule must add an ancestry identity with independent arithmetic content. Until that identity closes all four arms, both the rule and absolute boundedness remain open.

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